BA-TOS: Boundary-Aware Tree-of-Shapes Markers for 3D Grain Instance Segmentation in Micro-CT

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***Abstract*—Accurate instance segmentation of individual grains in three-dimensional micro-computed tomography (micro-CT) images is essential for particle-scale analysis, but marker-controlled watershed segmentation can fail when selected marker regions become structurally degenerate. This paper introduces BA-TOS, an interpretable boundary-aware marker-selection strategy based on the Tree of Shapes (ToS). The method extracts dark MIN-like ToS candidates, rejects candidates connected to the crop boundary, refines internal regions into markers, and applies a marker-controlled distance watershed. An image-derived safeguard detects voxel-scale collapse of exclusive ToS supports and selectively substitutes distance-transform markers without using ground truth. This safeguard is deliberately limited: it is activated only for the two crops exhibiting support collapse, while the remaining ten outputs are bitwise identical to the original ToS route. The core pipeline retains pooled F1@0.50 = 0.974, and the limited safeguard raises crop-macro mean F1@0.50 from 0.824 ± 0.367 to 0.906 ± 0.210. On one representative crop, F1 is unchanged for score thresholds 0.25–0.45 and NMS radii 6–10, whereas capacities below $K = 3600$ reduce performance. Nested experiments at two locations show that 160^3 crops remain close to 200^3, whereas 128^3 crops lose substantial spatial context. External validation across materials and full volumes remains necessary.**

***Index Terms*—3D instance segmentation, micro-computed tomography, Tree of Shapes, marker-controlled watershed, mathematical morphology.**

I. Introduction

Micro-computed tomography (micro-CT) enables non-destructive three-dimensional observation of granular materials during mechanical experiments. In this sense, instance segmentation of individual grains is a fundamental step for measuring particle morphology, displacement, contact evolution, and deformation-related structural changes. In [1], [2], geomechanics studies have emphasized that particle morphology and inter-particle friction strongly affect granular deformation processes, and in-operando X-ray tomography enables particle-scale observation and tracking throughout mechanical tests, confirming the need for reliable particle-wise segmentation.

Segmenting individual grains in 3D micro-CT volumes remains challenging due to dense packing, touching particles, boundary artifacts, acquisition artifacts, and local intensity variations. In [3], the grain motions in 3D CT sequences were studied based on topological analysis. More specifically, several morphology-based segmentation strategies were investigated to tackle this problem, including Min-tree core extraction [4], [5], watershed segmentation [6], Max-tree processing [4], [5], and Tree-of-Shapes-based approaches [7]–[9].

A seed-based watershed segmentation [6] is highly dependent on marker (or seed) quality. Markers based on local maxima of a distance transform may generate fragmented instances, while Min-tree-based core extraction [4], [5] may under-recover valid grains when markers are sparse or unstable. These limitations motivate marker extraction strategies that are geometrically meaningful, topologically informed, and robust to boundary-connected background structures. Moreover, in [3], dark nuclei are extracted by using Min-tree filtering; however, noise and scan imperfections introduce additional extrema, and watershed quality remains dependent on the markers and foreground support. To address these issues, we use the Tree of Shapes (ToS), a self-dual morphological hierarchy that represents the inclusion of level-line shapes and supports connected-operator processing [7]–[9]. Unlike a Min-tree, which represents only lower level sets, the ToS provides a single hierarchy in which candidate polarity, ancestry, boundary contact, and exclusive spatial support can be analyzed consistently. In the frozen configuration evaluated here, the retained watershed markers are MIN-like; therefore, we do not attribute the observed gains to MAX-like candidates. The contribution instead lies in boundary-aware selection from the ToS hierarchy and in the diagnosis of cases in which the selected ToS supports collapse to voxel-scale regions. A direct Min-tree comparison is included in the evaluation. Learning-based methods such as Cellpose [10] are treated as complementary rather than as state-of-the-art baselines because the present setting targets training-free, reproducible marker generation.

This article proposes BA-TOS, a boundary-aware ToS marker-selection strategy for 3D grain instance segmentation. The boundary-aware ToS route remains the core method. The revision does not replace or reformulate this route: it adds only a restricted, ground-truth-free diagnostic for a specific crop-level support-collapse failure mode. When the selected ToS supports indicate this degenerate regime, a deterministic safeguard changes only the marker source by using distance-transform markers inside the same foreground support; otherwise, the original ToS markers are retained. The main contributions are:

- a boundary-aware topological marker-selection strategy that rejects crop-boundary-connected dark candidates before watershed initialization;

Algorithm 1 BA-TOS marker selection with a structural safeguard.

Input grayscale crop $I : \Omega \to \mathbb{R}$
Output instance label volume L

Compute conservative flattened crop I_f.
Extract dark MIN-like ToS candidates $\mathcal{C}$.
Reject $\mathcal{C}_b = \{C \in \mathcal{C} \mid C \cap \partial\Omega \neq \emptyset\}$.
Build labeled internal ToS markers S_{ToS}.
Assign each leaf voxel to its nearest selected ToS ancestor and obtain exclusive region volumes $\{v_i\}$.
$g \leftarrow [\mathrm{median}(v_i) \leq 3] \wedge [Q_{0.75}(v_i) \leq 4]$.
Compute Otsu foreground mask M and distance transform D.
$S \leftarrow S_{\mathrm{EDT}}$ if g; otherwise $S \leftarrow S_{\mathrm{ToS}}$.
Compute $L = \mathrm{Watershed}(-D, S, M)$.
Remove predicted instances smaller than τ_{size} and return L.

- a controlled comparison with Min-tree-core and EDT-Otsu marker sources under the same object-level evaluation protocol;
- a ground-truth-free structural diagnostic and deterministic fallback safeguard for voxel-scale ToS-support collapse;
- crop-macro robustness, component ablation, core-parameter sensitivity, crop-extent sensitivity, physical threshold interpretation, and explicit limits on dataset scale and full-volume generalization.

II. A Boundary-Aware Tree-of-Shapes-Based Marker Selection Strategy

BA-TOS takes a 3D grayscale micro-CT crop as input and produces an instance-labeled volume. Its core principle remains unchanged: watershed basins should be initialized from internal grain markers rather than from boundary-connected or inter-grain dark structures. The structural safeguard is a narrow marker-source exception, not a separate segmentation method: conservative flattening, boundary rejection, the foreground mask, watershed relief, connectivity, and post-filtering remain shared. The pipeline comprises conservative flattening, ToS candidate extraction, boundary rejection, internal-marker construction, structural-degeneracy diagnosis, selective marker routing, distance watershed, and size-aware filtering. We denote the ToS-only route by BA-TOS$_{\mathrm{ToS}}$ and the final selective variant by BA-TOS$_{\mathrm{sel}}$.

A. Conservative volumetric flattening

Given an input 3D grayscale crop $I : \Omega \to \mathbb{R}$, a conservative intensity-flattening step is first applied to attenuate low-frequency illumination and beam-hardening variations while preserving local grain-scale structures. Let B_σ denote a smooth background estimate obtained from I at a spatial scale larger than the expected grain radius. The flattened image is defined as

$$I_f(x) = I(x) - B_\sigma(x) + \mathrm{median}_{y \in \Omega} B_\sigma(y), \quad x \in \Omega. \quad (1)$$

The term conservative means that the background correction is only used to reduce slow intensity drift; it does not threshold the image, remove connected components, or impose a foreground mask. The normalized volume I_f therefore denotes an intensity-corrected grayscale volume used as input for candidate extraction, not a binary segmentation or probability map.

B. Tree-of-Shapes candidate extraction

From the flattened volume I_f, dark candidate regions are extracted using ToS-based connected-operator processing. The ToS is not used as the final segmentation; its nodes provide a hierarchical candidate representation for marker selection and spatial auditing. Although the hierarchy is self-dual, the frozen internal eligibility rule retains MIN-like supports in the evaluated data. Consequently, the MIN and dual downstream marker configurations are identical; no empirical gain is attributed to MAX-like candidates.

For each MIN-like ToS component C, the saliency score is

$$s(C) = h(C)\, a(C), \quad (2)$$

where $h(C)$ is the normalized parent–child contrast and $a(C)$ measures consistency with the admissible marker-size interval. Candidates satisfy

$$A_{\min} \leq |C| \leq A_{\max}, \qquad s(C) \geq \tau_s. \quad (3)$$

The frozen configuration uses $\tau_s = 0.35$, a total candidate-pool capacity of $K = 3600$, and a six-voxel non-maximum-suppression radius. In the self-dual selection stage, this capacity comprises up to 1800 MIN-like and 1800 MAX-like nodes before internal-support eligibility and spatial rejection. Thus, K is not the final marker count; the evaluated crops contain 53–405 retained ToS markers, and the downstream eligible markers are MIN-like. All values are held fixed across both scans and are not tuned per crop. Section III-D evaluates $\tau_s \in [0.25, 0.45]$, $K \in \{900, 1800, 3600, 7200\}$, and NMS radii from 4 to 10 voxels.

C. Boundary-connected rejection

Not all dark candidates correspond to internal grain markers. Some candidates are connected to the crop boundary and represent external background or inter-grain regions. BA-TOS explicitly separates boundary-connected dark structures from internal candidates. Let $\mathcal{C}$ be the set of candidate connected components. A component $C \in \mathcal{C}$ is rejected if:

$$C \cap \partial\Omega \neq \emptyset, \quad (4)$$

where $\partial\Omega$ denotes the boundary of the 3D crop domain. In practice, this rule rejects any candidate that touches one of the six faces of the 3D crop. Figure 2 illustrates this marker-refinement step by separating boundary-connected dark candidates from internal candidates retained for watershed initialization.

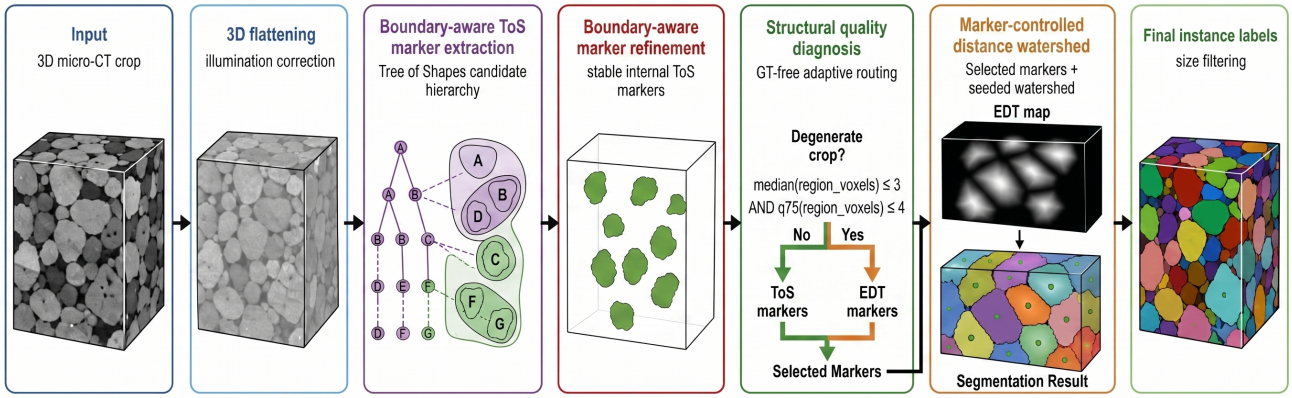


Fig. 1. Overview of the BA-TOS pipeline. Internal ToS markers are preserved when their exclusive supports remain structurally admissible; an image-derived diagnostic routes only voxel-scale support-collapse cases to EDT-Otsu markers before the same marker-controlled watershed.

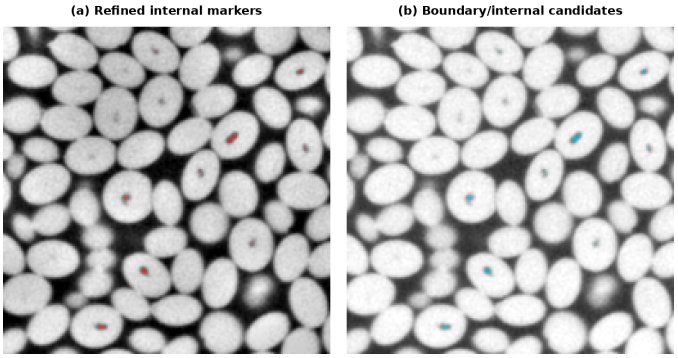


Fig. 2. Boundary-aware marker refinement. The boundary-aware step separates dark candidates connected to the crop boundary from internal dark candidates. This reduces markers placed in background or inter-grain regions and preserves markers that are more suitable for initializing the watershed.

D. Internal marker refinement

The remaining candidates are converted into labeled 3D markers after removing boundary-connected and spatially inadmissible regions. Each surviving candidate is represented by a unique marker label; this stage produces a marker image, not a final grain segmentation.

E. Structural marker-support diagnostic

Let $\mathcal{N} = \{n_i\}$ be the selected ToS nodes. To avoid counting overlapping ancestors multiple times, each leaf voxel is assigned to its nearest selected ancestor, defining an exclusive support R_i and volume $v_i = |R_i|$. We compute

$$m_v = \mathrm{median}_i(v_i), \qquad q_v = Q_{0.75}(\{v_i\}), \tag{5}$$

and activate the fallback safeguard when

$$g = (m_v \le 3) \wedge (q_v \le 4). \tag{6}$$

If g is false, BA-TOS preserves S_{ToS}. If g is true, the safeguard uses S_{EDT}, obtained from local maxima of the Euclidean distance transform inside the same cleaned Otsu mask. No reference labels, IoU values, or crop identities enter this decision. The rule was frozen after retrospective failure analysis and is not adjusted during evaluation. In the present twelve-crop evaluation, it activates only for EFRGP01_00_crop05 and EFRGP01_01_crop05; for the other ten crops, the original ToS markers and final segmentations are preserved exactly.

TABLE I
FROZEN PARAMETERS AND THEIR OPERATIONAL ROLES.

Parameter	Value	Purpose
Crop / voxel pitch	200^3 / 100 μm	Evaluation scale
Flattening	$\lambda_{xy} = 15$	Low-frequency correction
ToS area factor / range	22 / 2–2200	Candidate filtering
ToS score threshold	0.35 (tested 0.25–0.45)	Saliency rejection
Candidate capacity	3600 (tested 900–7200)	Pre-spatialization upper bound
Spatial NMS radius	6 (tested 4–10) voxels	Suppress nearby candidates
Boundary rule	$C \cap \partial\Omega \neq \emptyset$	Reject external candidates
Gate statistics	$m_v \le 3$, $q_v \le 4$	Detect degenerate ToS support
Fallback	EDT-Otsu markers	Gate-positive crops only
Mask / closing	Otsu offset -10 / 1 voxel	Foreground support
Prediction / valid GT	500 / 1000 voxels	Fragment / validity filters
IoU thresholds	0.25, 0.50, 0.75	Multi-threshold matching

F. Marker-controlled distance watershed

A foreground mask M is obtained by Otsu thresholding of I_f, followed by connected-component cleanup. The Euclidean distance transform D is computed inside M, and the selectively chosen marker image S initializes

$$L = \mathrm{Watershed}(-D, S, M). \tag{7}$$

Both routes use the same mask, relief, connectivity, and post-filtering; therefore, the gate changes only marker provenance.

G. Size-aware post-filtering

Predicted labels with fewer than 500 voxels are removed, while reference instances smaller than 1000 voxels are excluded from object-level evaluation. The scans have a voxel pitch of approximately $p = 100\,\mu$m/voxel [1]. For a sphere-equivalent diameter

$$d_{\mathrm{eq}}(V) = 2\left(\frac{3V}{4\pi}\right)^{1/3} p, \tag{8}$$

500 and 1000 voxels correspond to approximately 0.985 mm and 1.241 mm, respectively. Thus, the prediction filter removes sub-grain fragments below the minimum physically admissible reference scale rather than matching the evaluation cutoff exactly.

III. Experimental Setup

A. Dataset

Experiments use two consecutive scans from the public EFRGP01 4D X-ray tomography sequence, EFRGP01_00 and EFRGP01_01 [1], [2]. The reconstructed voxel pitch is approximately $100\,\mu$m. Six representative $200\times200\times200$ crops are evaluated per scan, totaling twelve crops. Reference labels are used only for quantitative evaluation; particles touching a crop face or containing fewer than 1000 voxels are excluded from the valid-reference set. The two scans test temporal consistency under small deformation changes, but they do not constitute external validation across materials or acquisition protocols.

B. Compared methods

We compare four training-free marker pipelines:

- **Min-tree-core+WS**: Min-tree core extraction followed by seeded watershed [3];
- **Otsu+WS**: cleaned Otsu foreground support with distance-transform markers and watershed [11];
- **BA-TOS**$_{\mathrm{ToS}}$: the boundary-aware ToS route without fallback;
- **BA-TOS**$_{\mathrm{sel}}$: the proposed selective route in Algorithm 1.

The direct Min-tree comparison addresses whether a dark-only component hierarchy is sufficient. Higra provides the hierarchical and watershed primitives [12]; no method is trained on the reference labels.

C. Evaluation protocol

Object matching uses IoU thresholds of 0.25, 0.50, and 0.75. Predictions seeded exclusively inside reference components excluded by the boundary/size validity rules are ignored; background-seeded predictions remain false positives. This ignore-aware protocol prevents invalid border particles from distorting the comparison. For matched objects, we set

$$P=\frac{TP}{TP+FP},\qquad R=\frac{TP}{TP+FN},\qquad F1=\frac{2PR}{P+R}. \tag{9}$$

The primary robustness summary reports the mean, standard deviation, median, and minimum of per-crop F1. This crop-macro view prevents a large number of easy objects from hiding catastrophic crop-level failures. Pooled object counts are retained only for the component ablation. The oversegmentation ratio is $N_{\mathrm{pred}}/N_{\mathrm{gt}}$.

D. Sensitivity analyses

We performed two complementary sensitivity experiments. First, a one-factor-at-a-time analysis on the representative 200^3 crop EFRGP01_00_crop04 varied the ToS score threshold $\tau_s \in \{0.25, 0.30, 0.35, 0.40, 0.45\}$, the candidate capacity $K \in \{900, 1800, 3600, 7200\}$, and the NMS radius $r \in \{4, 6, 8, 10\}$ voxels, while freezing every other stage. Second, crop-extent sensitivity was evaluated on nested central cubes of 128^3, 160^3, and 200^3 voxels at two spatial locations, retaining parent-scale parameters. The primary crop-size comparison uses the seed-ignore protocol defined above. A complementary strict boundary-margin audit over the twelve benchmark crops was also generated; because increasing the margin changes the set of evaluable reference objects, it is treated as a protocol diagnostic rather than as evidence of algorithmic invariance.

TABLE II
Crop-macro robustness across the twelve 3D crops.

Method	Mean F1@.50	SD	Median	Minimum	Mean F1@.75
Min-tree-core+WS	0.768	0.315	0.890	0.088	0.751
Otsu+WS	0.830	**0.182**	0.882	**0.260**	0.770
BA-TOS$_{\mathrm{ToS}}$	0.824	0.367	**0.985**	0.012	0.817
BA-TOS$_{\mathrm{sel}}$	**0.906**	0.210	**0.985**	**0.260**	**0.899**

TABLE III
Pooled core-pipeline ablation at IoU@0.50.

Variant	P@.50	R@.50	F1@.50	Overseg.	FP	FN
BA-TOS core	0.980	0.967	0.974	0.987	39	77
w/o boundary rejection	0.000	0.000	0.000	0.034	102	2808
w/o size filter	0.845	0.967	0.881	1.331	618	77
w/o flattening	0.705	0.662	0.672	0.688	70	706

IV. Results

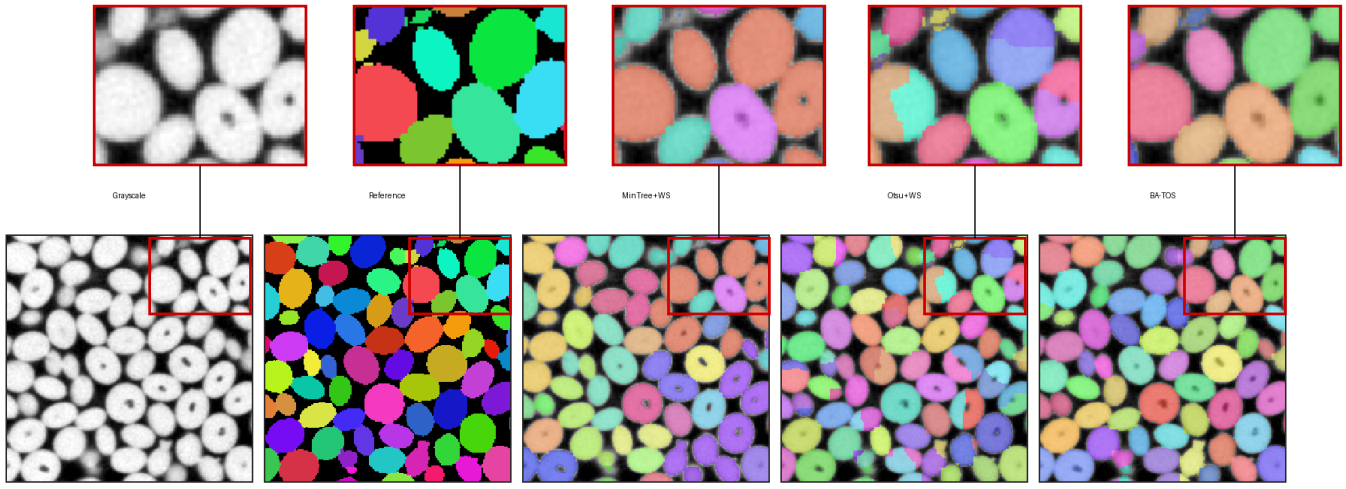
A. Quantitative comparison and failure recovery

Table II exposes an important distinction between typical and worst-case behavior. The safeguard is therefore interpreted as limited recovery of a diagnosed failure mode, not as a replacement or reformulation of the core BA-TOS route. The pooled core-pipeline result remains F1@0.50 = 0.974, as reported in Table III; however, pooling objects across crops hides severe crop-level failures. In the crop-macro analysis, BA-TOS$_{\mathrm{ToS}}$ has a high median F1@0.50 of 0.985 but a minimum of 0.012. The structural safeguard activates only on EFRGP01_00_crop05 and EFRGP01_01_crop05, whose exclusive ToS-region statistics are $m_v = 2$ and $q_v = 3$. Routing these two crops to EDT-Otsu increases mean F1@0.50 from 0.824 to 0.906, increases mean F1@0.75 from 0.817 to 0.899, and reduces the F1@0.50 standard deviation from 0.367 to 0.210. The other ten segmentations are bitwise identical to the frozen ToS outputs. Because Table II is crop-macro and Table III is pooled, their F1 values are complementary but not directly comparable.

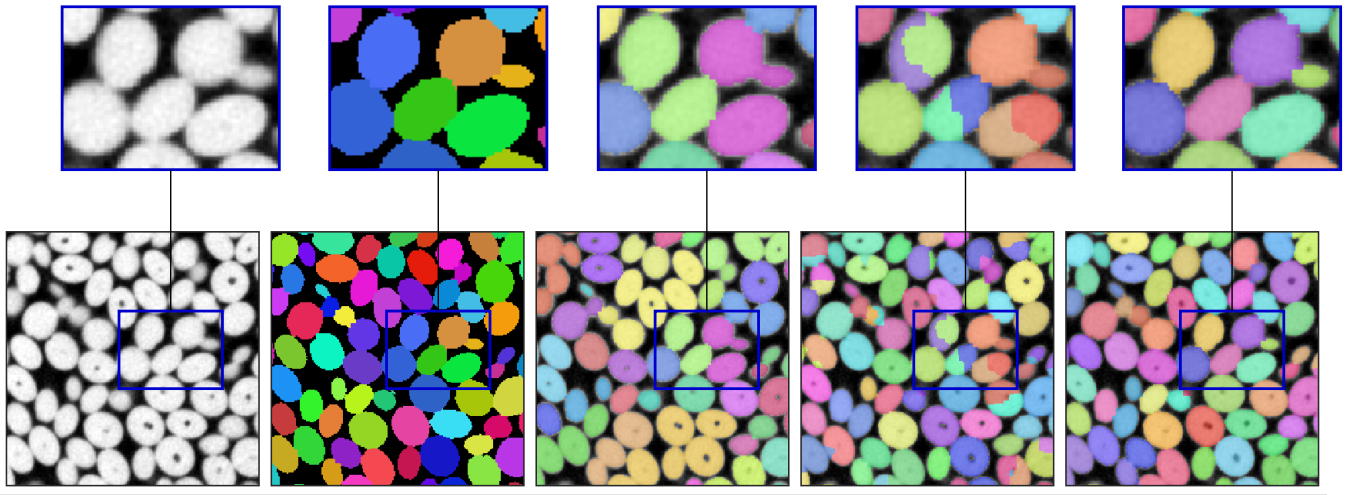
Table III retains the pooled component ablation because it isolates the mechanisms of the core marker pipeline. Removing boundary rejection collapses IoU@0.50 matching, confirming that unfiltered hierarchical candidates are not valid watershed markers. Removing the size filter preserves recall but increases false positives, while removing flattening increases false negatives. The qualitative connection is visible in Fig. 3: Min-tree-core+WS misses grains because its cores are sparse, whereas Otsu+WS produces fragmented labels because distance maxima over-seed touching regions. The selective gate does not replace these components; it acts only when the already boundary-filtered ToS regions collapse to voxel-scale supports.

B. Structural-safeguard stability

The two triggered crops have $m_v = 2$ and $q_v = 3$, whereas the closest non-triggered crop has $m_v = 17$ and $q_v = 43$. Across an integer grid, 2230 threshold pairs produce exactly



(a) EFRGP01_00_crop07



(b) EFRGP01_01_crop07

Fig. 3. Qualitative comparison on two gate-negative crops. Each panel shows the grayscale slice, reference labels, Min-tree-core+WS, Otsu+WS, and the BA-TOS ToS route. Min-tree markers under-recover grains, Otsu distance markers fragment touching particles, and boundary-aware ToS markers produce more coherent instances.

TABLE IV
ONE-FACTOR-AT-A-TIME SENSITIVITY ON EFRGP01_00_CROP04.

Factor	Tested values	F1@.50	F1@.75
τ_s	.25/.30/.35/.40/.45	.985/.985/.985/.985/.985	.978/.978/.978/.978/.978
K	900/1800/3600/7200	.510/.833/.985/.998	.478/.809/.978/.998
NMS	4/6/8/10	.976/.985/.985/.985	.965/.978/.978/.978

the same two-crop routing; the primary rule lies inside the rectangle $2 \le m_{\lim} \le 16$ and $3 \le q_{\lim} \le 42$. Leave-one-out removal and 5000 subsamples without replacement at 90%, 80%, and 70% preserve every crop decision. An ordinary with-replacement bootstrap yields 0.9744 gate probability for one discrete 53-marker crop, below the predeclared 0.99 criterion; we therefore report finite-sample stability but do not claim population-level or external validation.

C. Core-parameter and crop-extent sensitivity

Table IV shows unchanged F1 across $\tau_s = 0.25$–0.45 and NMS radii 6–10; radius 4 changes F1@0.50 by -0.009. Capacity is consequential: $K = 900/1800/3600/7200$ yields F1@0.50 of 0.510/0.833/0.985/0.998. Hence, $K = 3600$ is a sufficient frozen operating point, not a claim of capacity invariance.

TABLE V
CROP-EXTENT SENSITIVITY AT TWO LOCATIONS (F1@0.50).

Size	EFRGP01_00	EFRGP01_01	Mean	Mean Δ vs. 200^3
128^3	0.830	0.871	0.851	-0.134
160^3	0.979	0.955	0.967	-0.018
200^3	0.985	0.985	0.985	0.000

Table V shows mean F1@0.50 of 0.851, 0.967, and 0.985 for 128^3, 160^3, and 200^3. Thus, 160^3 remains close to the parent scale, whereas 128^3 loses substantial context. In the strict twelve-crop margin audit, mean F1@0.50 decreases from 0.692 at margin 0 to 0.523 at margin 12 while evaluable references decrease from 234.0 to 148.6; because the reference set changes, this diagnoses protocol sensitivity rather than algorithmic invariance. Both sensitivity studies remain descriptive and limited in scale.

Exploratory learning-based transfer.: As an exploratory analysis rather than a direct baseline, Cellpose3D trained on same-crop BA-TOS pseudo-labels obtained recall 0.995 but precision 0.472 because of severe oversegmentation. This result reinforces that foreground localization alone does not replace reliable instance-marker control.

D. Visual comparison

Figure 3 compares grayscale slices, reference labels, Min-tree-core+WS, Otsu+WS, and the ToS route on two gate-negative crops. Min-tree-core+WS under-recovers grains because sparse cores do not initialize all valid basins. Otsu+WS introduces fragmented instances around touching particles because multiple distance maxima compete inside the same grain-scale support. BA-TOS produces more coherent labels by rejecting boundary-connected candidates before watershed initialization. The two gate-positive crop05 cases are summarized quantitatively because the revised routing changes only their marker source.

V. Discussion

The crop-level results clarify both the strength and the limitation of the ToS route. In ten crops, boundary-aware ToS markers provide F1@0.50 between 0.950 and 0.991. In two crop05 cases, however, the selected exclusive regions collapse to a median of two voxels, producing too few or duplicated grain seeds. Repositioning a marker inside the same degenerate region cannot recover missing regions; the relevant signal is therefore crop-level marker-support geometry, not merely marker centrality.

The direct Min-tree comparison answers why a ToS hierarchy is retained even though the frozen markers are MIN-like. We do not claim an empirical contribution from MAX-like nodes. The ToS supplies a unified inclusion hierarchy from which parent–child contrast, boundary contact, ancestry, and exclusive support are computed consistently. Min-tree-core+WS attains lower macro F1 and lower worst-case performance, while raw ToS candidates also fail without the proposed boundary constraint. Thus, the contribution is the spatial/topological marker-selection rule and its failure diagnosis, not a new ToS construction.

With $100\,\mu$m voxels, the 500- and 1000-voxel cutoffs correspond to sphere-equivalent diameters of approximately 0.985 and 1.241 mm; the much smaller diagnostic thresholds identify voxel-scale supports, not physical grains. Although routing is stable over a broad threshold rectangle, the safeguard was formulated retrospectively and requires independent validation.

The sensitivity study shows local plateaus for the score threshold and NMS radius, but K must prevent hierarchy under-sampling. We retain $K = 3600$ because the gain at 7200 is small while computational cost increases. This one-crop, one-factor-at-a-time experiment does not quantify parameter interactions.

Crop extent is a genuine limitation: 160^3 remains close to 200^3 at both locations, whereas 128^3 loses mean F1@0.50 of approximately 0.134. A full-volume implementation should use overlapping tiles with a grain-radius halo, apply boundary rejection at the outer halo, retain instances by core-centroid ownership, and reconcile overlaps by one-to-one IoU matching. This workflow is unvalidated. Current evidence remains limited to two consecutive scans of one material, two crop-size locations, and one local parameter study; it does not establish full-volume or cross-material generalization.

VI. Conclusion

This paper presented BA-TOS, a boundary-aware ToS marker-selection strategy for 3D micro-CT grain instances. The core ToS route remains unchanged; a restricted ground-truth-free safeguard changes only the marker source in the two detected support-collapse cases, increasing crop-macro mean F1@0.50 from 0.824 to 0.906 while preserving the other ten segmentations exactly. Local sensitivity analysis shows stable score-threshold and NMS ranges but degradation at insufficient candidate capacity. Nested crops show performance near 200^3 at 160^3, but substantial context loss at 128^3. The safeguard is retrospective, the sensitivity studies are limited, and validation across materials and full volumes remains necessary.

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